

UAE REFINARIES

1. Project Overview and Objective

This project involves cleaning, transforming, and analysing raw data using **Excel** and creating an interactive **Power BI dashboard** to derive meaningful business insights.

The main objective is to demonstrate data pre-processing techniques using Excel and an interactive Power BI dashboard visualization to make informed decisions.

2. Data Sources

- **Source Description and Timeline:** Refinery capacity distribution over UAE Emirates and 1975-2025.
- **Domain:** Oil & Gas

3. Problem Statement

- To analyze refinery performance across UAE Emirates and identify top-performing regions.
- To study how refinery age, capacity, and crude oil prices affect production and profit.
- To compare petrol, diesel, and gas output among refineries to find efficiency trends.
- To evaluate employee count and cost to understand their impact on refinery operations.
- To use the findings to help improve planning and decision-making in the oil & gas sector.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Refinery Name	String	Name of Refinery.
Emirate	String	The UAE Emirate where the refinery is located.
Operator	String	The organization managing the refinery.
Capacity_BPD	Numeric	Crude oil in Barrels which refinery processed daily.
Commissioned Year	Numeric	The year the refinery started operations.

N0_of Employees	Numeric	Total number of employees working in refinery.
Crude Source	Categorical	Origin of crude oil for all refineries.
Annual_Throughput_Million_Tons	Numeric	Annual output of crude oil from refinery.
Environmental Rating	Numeric	Refineries environmental rating and Standards.
Petrol_Output_BPD	Numeric	Quantity of Petrol extract from crude oil in barrel per day.
Diesel_Output_BPD	Numeric	Quantity of Diesel extract from crude oil in barrel per day.
Gas_Output_BPD	Numeric	Quantity of Gas extract from crude oil in barrel per day.
Employee Category	Categorical	Role of each employee working in refinery.
Average Salary	Numeric	Average Salary of employees per day.
Output Productivity/Day	Numeric	Productivity of total number of employees in refinery per day.
Employee Productivity Level	Categorical	Based on the employee productivity classified as “Low”, “Medium” & “High”.
Salary/Barrel	Numeric	Salary of employees per barrel on daily basis.
Petrol Efficiency	Percentage	Total quantity of petrol extract from the crude oil in percentage.
Diesel Efficiency	Percentage	Total quantity of Diesel extract from the crude oil in percentage.
Gas Efficiency	Percentage	Total quantity of Gas extract from the crude oil in percentage.
Clean Energy	Categorical	Based on the environmental refineries are classified as “Good”, “Excellent” & “Poor”.
Refinery Age	Numeric	The number of years since the refinery was commissioned.
Cost/Barrel	Numeric	Cost of crude oil per barrel in AED

Revenue/Day	Numeric	Total Revenue of refineries per day.
Cost of Crude Oil as of each year	Numeric	Average cost of crude from the year 1975-2025.
Employee ID	Numeric	Unique identification number assigned to each employee working in the refineries.

5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel / Power Query)

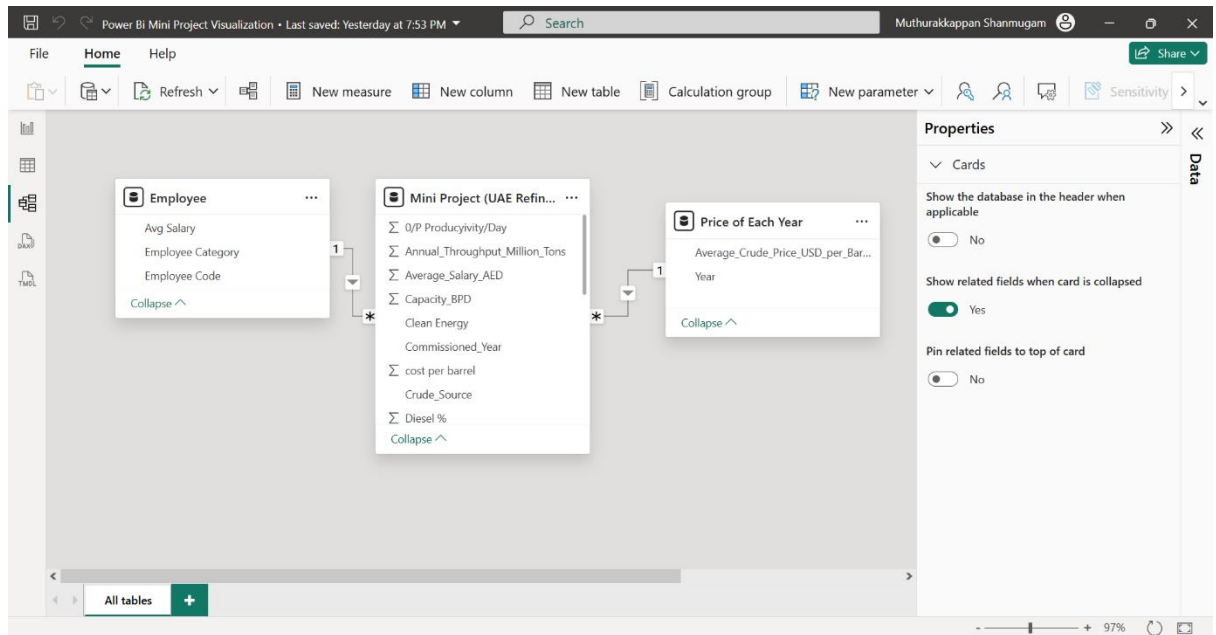
Tasks Performed:

- **Data Cleaning & Transformation:**
Created the calculated fields to find Productivity of employees per day, Employees salary per barrel, Efficiency of Petrol, Diesel and Gas, profit per barrel, Refinery Age & Revenue.
- **Filtering & Sorting:** Data was filtered to find refineries overall productivity in each emirate.
- **Pivot Tables:** Generated Pivot Tables for data summarization and initial insights.
 1. Create Pivot Tables for top 10 refineries.
 2. Create Pivot table to analyse productivity by crude sources.
- Fact Table are converted to three Dimensional Table

7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.
- **Relationship created for main table and dimensional table**

SCREENSHOT OF DATA MODELLING



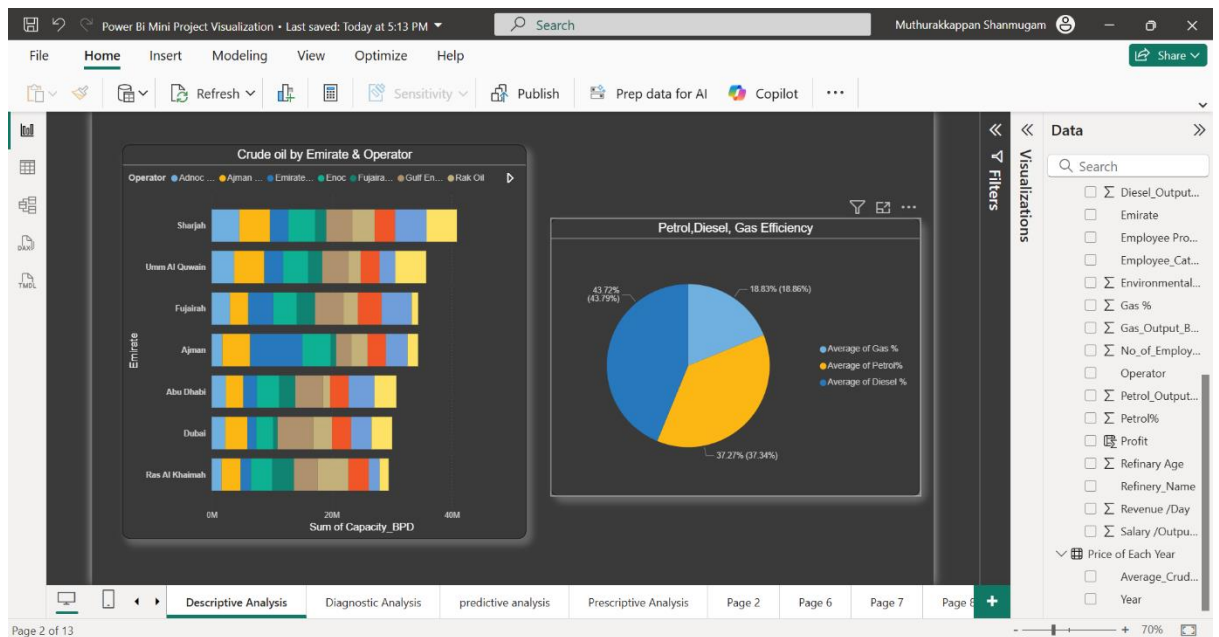
- **Calculated Columns & DAX Measures:** Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals.
 1. Created new column for profit calculation.
 2. Used Dax formula to identify total revenue per day, total crude oil per day, summarize quantity of crude oil in different refinery, calculate function to identify revenue in each emirates.

8. Analysis and Visualizations (Power BI)

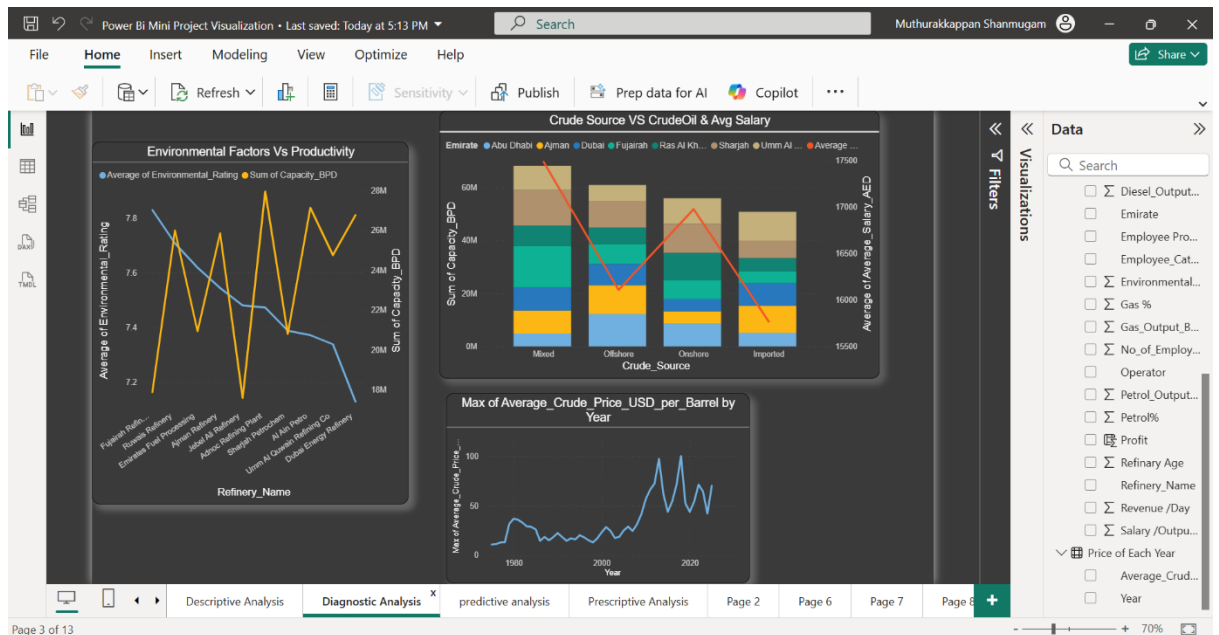
Dashboard Features:

- **Multiple Visualizations based on problem statement:** Bar charts, line charts, pie charts, cards, and tables to communicate insights.
 1. Bar chart for crude oil BPD VS Emirates & Operator.
 2. Pie chart for gas, petrol & diesel efficiency.
 3. Combo Chart for Refinery VS Age & its Efficiency.
 4. Narrative cards to used to communicate insights ie High productivity refineries & its Operator.
 5. Multiple Cards to specify details of high productivity refineries productivity of crude oil, extracted Oil, Gas and Petrol & profit.
- **Drill-down** are used to represent details of operator with its refineries and emirates.
- **Filters and slicers**
- **Created a consolidated Report /Dashboard.**

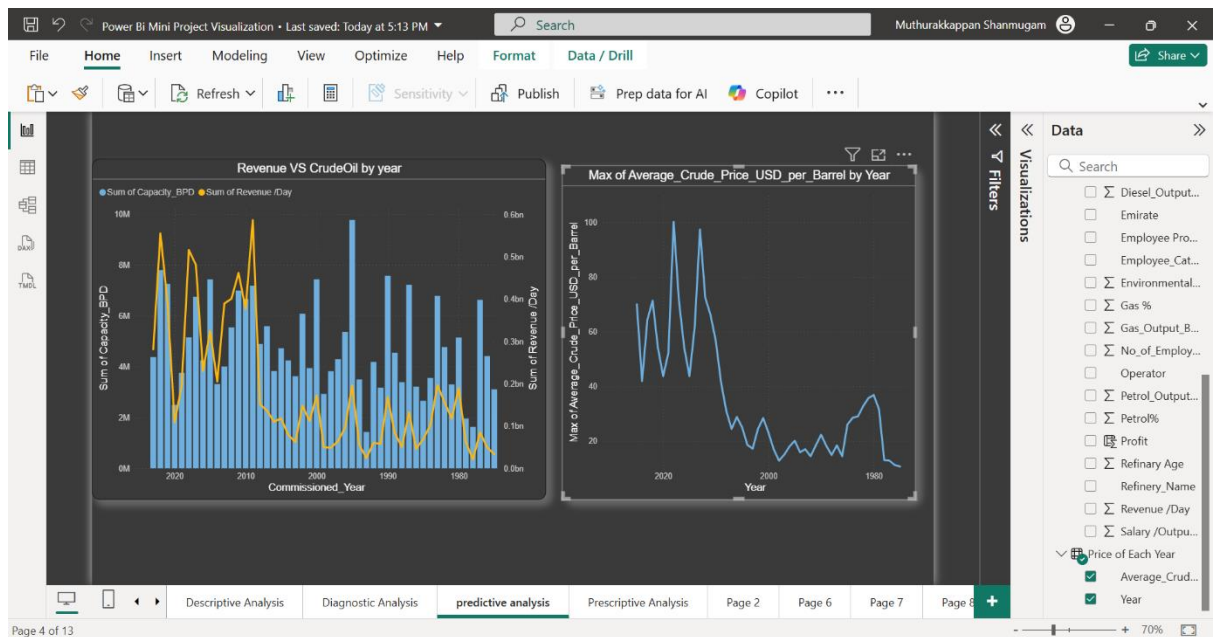
Descriptive Analysis



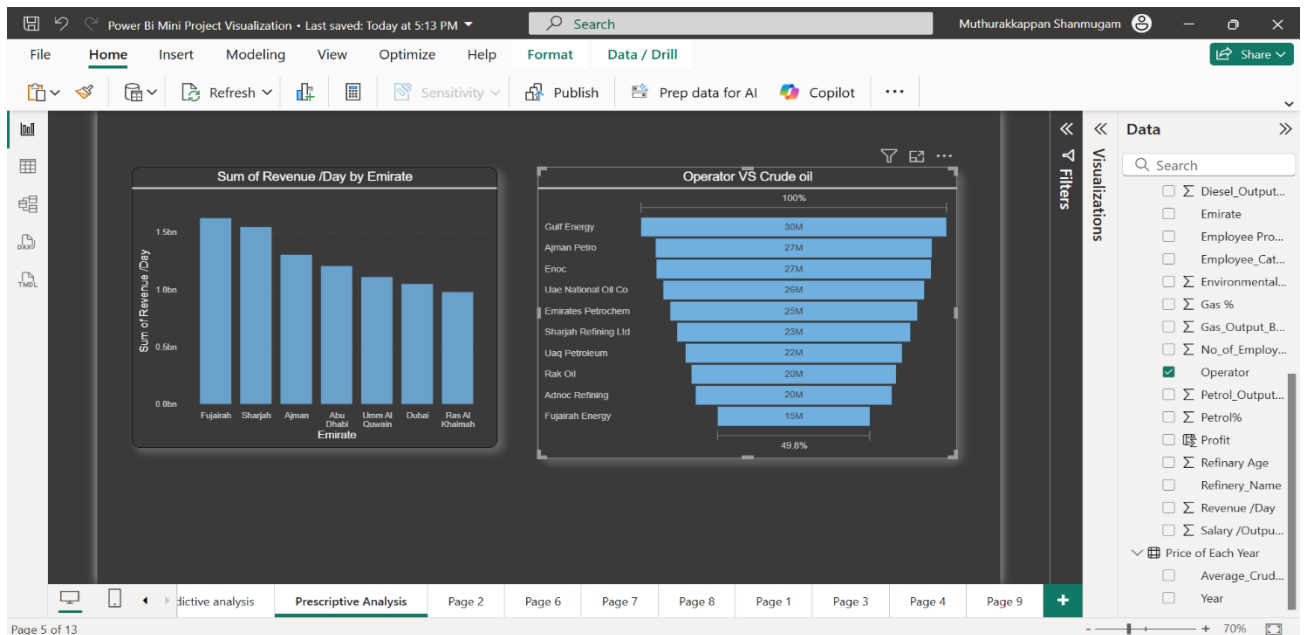
Diagnostic Analysis



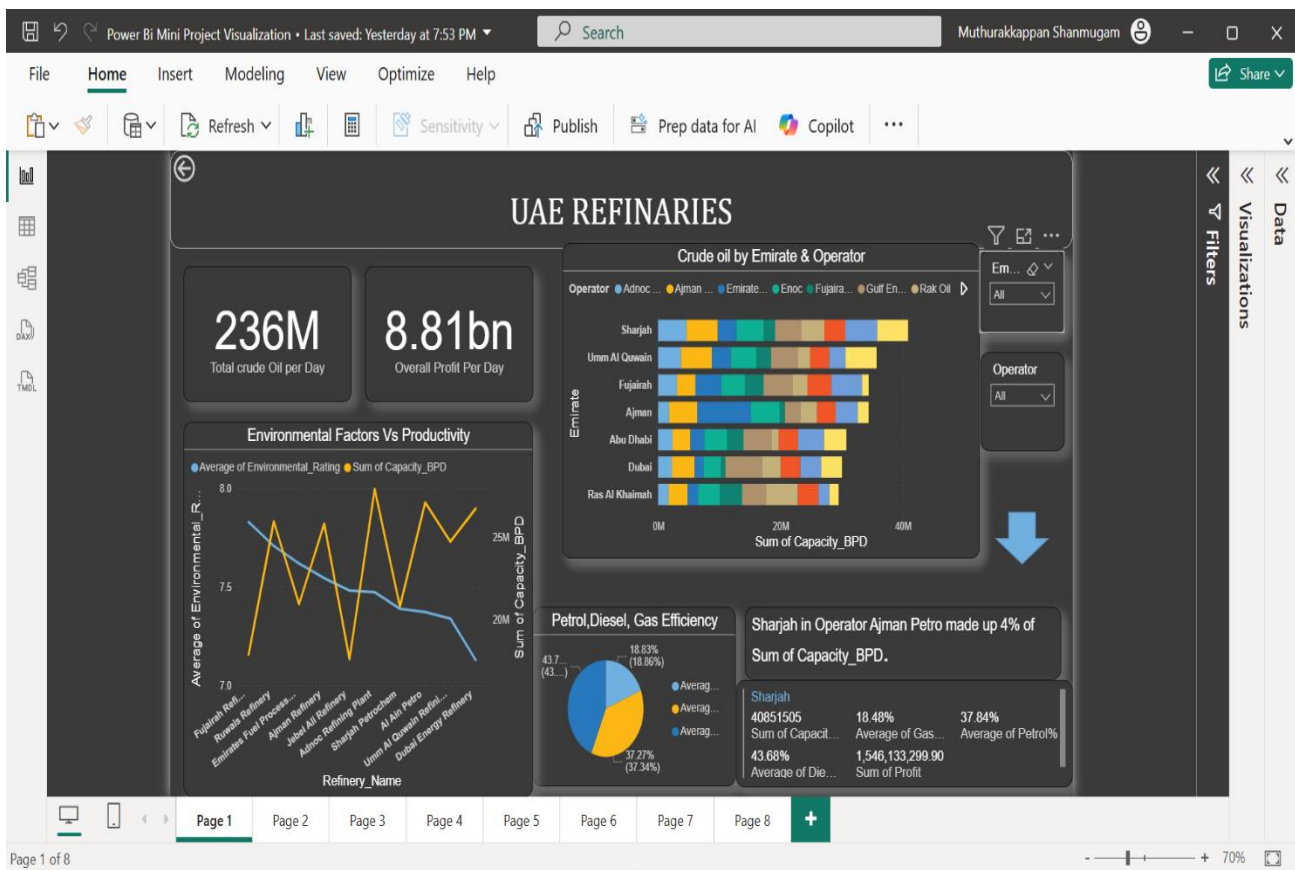
Predictive Analysis



Prescriptive Analysis



Consolidated Refinery Insights Report



9. Insights & Conclusions

- **Key Findings:**
- Sharjah Refinery recorded the **highest overall production capacity and revenue**, contributing approximately **19% of the total refining output** in the dataset.
- The overall trend shows **steady growth in refining capacity and profit** with minor dips corresponding to fluctuations in global crude oil prices.

- **Analysis insight**

EMIRATES	Total Capacity BPD
Sharjah	40851505
Um Al Quwain	35718228
Fujairah	34402496
Ajman	34393102
Abu Dhabi	30763738
Dubai	30040449
Ras Al Khaimah	29495496

- Efficiency of Diesel is higher than the Petrol and Gas.
- Um Al Quwain Refineries has highest revenue because major refineries are Onshore
- Renovate low-output refineries.
- Invest in top 3 high-margin operators.
- Upgrade technology for low-rated plants and Expand capacity and support top-performing operators to boost daily profit.

10. Conclusions

The combined analysis in Excel and Power BI revealed that UAE refineries achieve a total crude capacity of **236M BPD** and a daily profit of **AED 8.81B**. While high-performing plants drive most of the profit, lower-rated refineries show potential for improvement through technology upgrades and optimized crude distribution. Overall, the study highlights opportunities to enhance refinery efficiency, profitability, and environmental sustainability across the UAE.